Supplementary Information

Poly(A) tail retention of LINE-1 transcripts under arsenite stress covaries with m^6A density and structural completeness

Supplementary Figure S1: L1 Expression Landscape

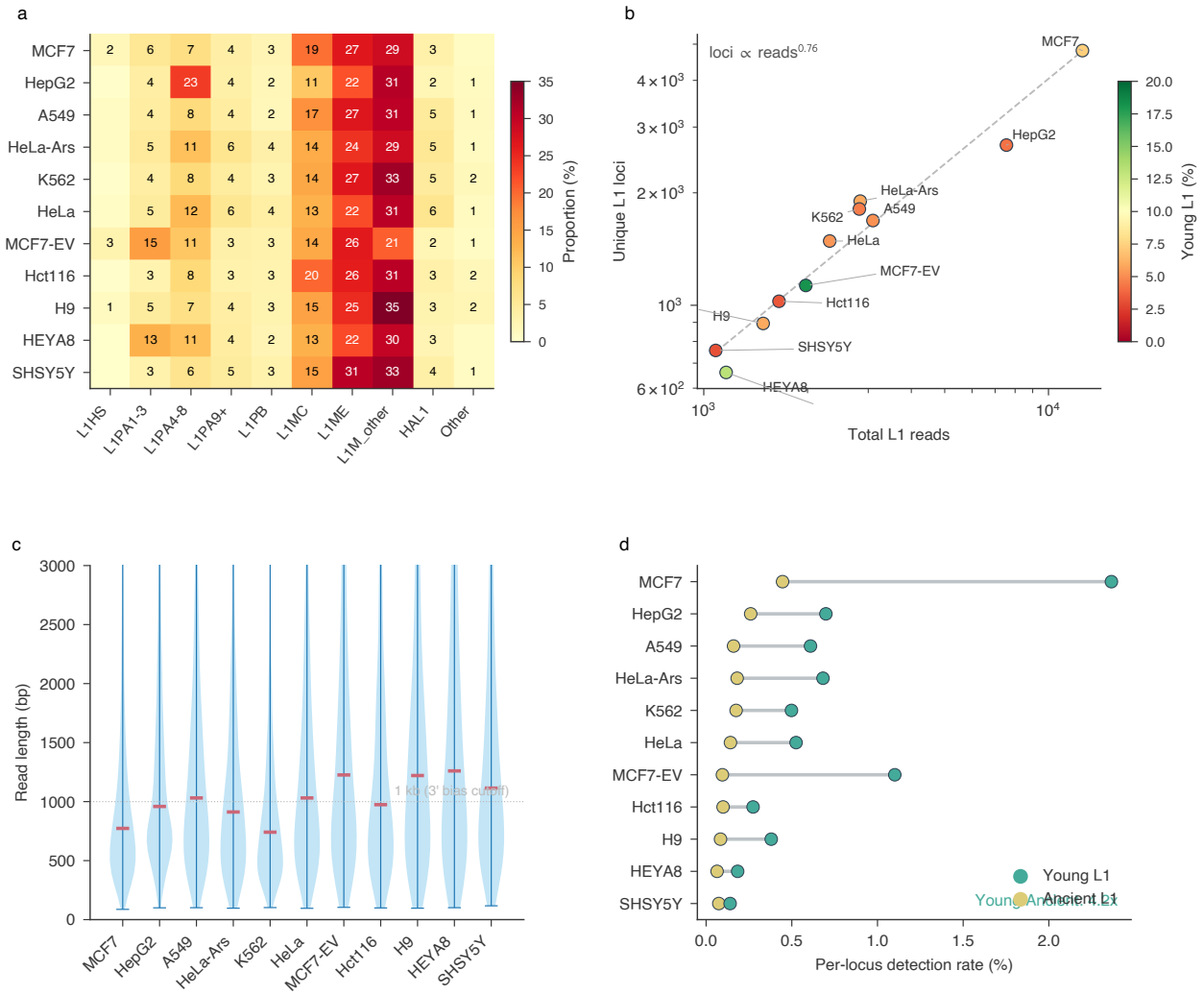

Figure 1: **L1 expression landscape across nine human cell lines.** **(a)** Heatmap of L1 subfamily composition. Ancient elements dominate across all cell lines (63–79%). **(b)** Total L1 reads vs. unique loci (log-log scale). **(c)** Violin distributions of L1 read lengths per cell line. **(d)** Per-locus detection rates for young vs. ancient L1.

Supplementary Figure S2: Exon-Embedded L1 and Baseline m^{6}A–Poly(A)

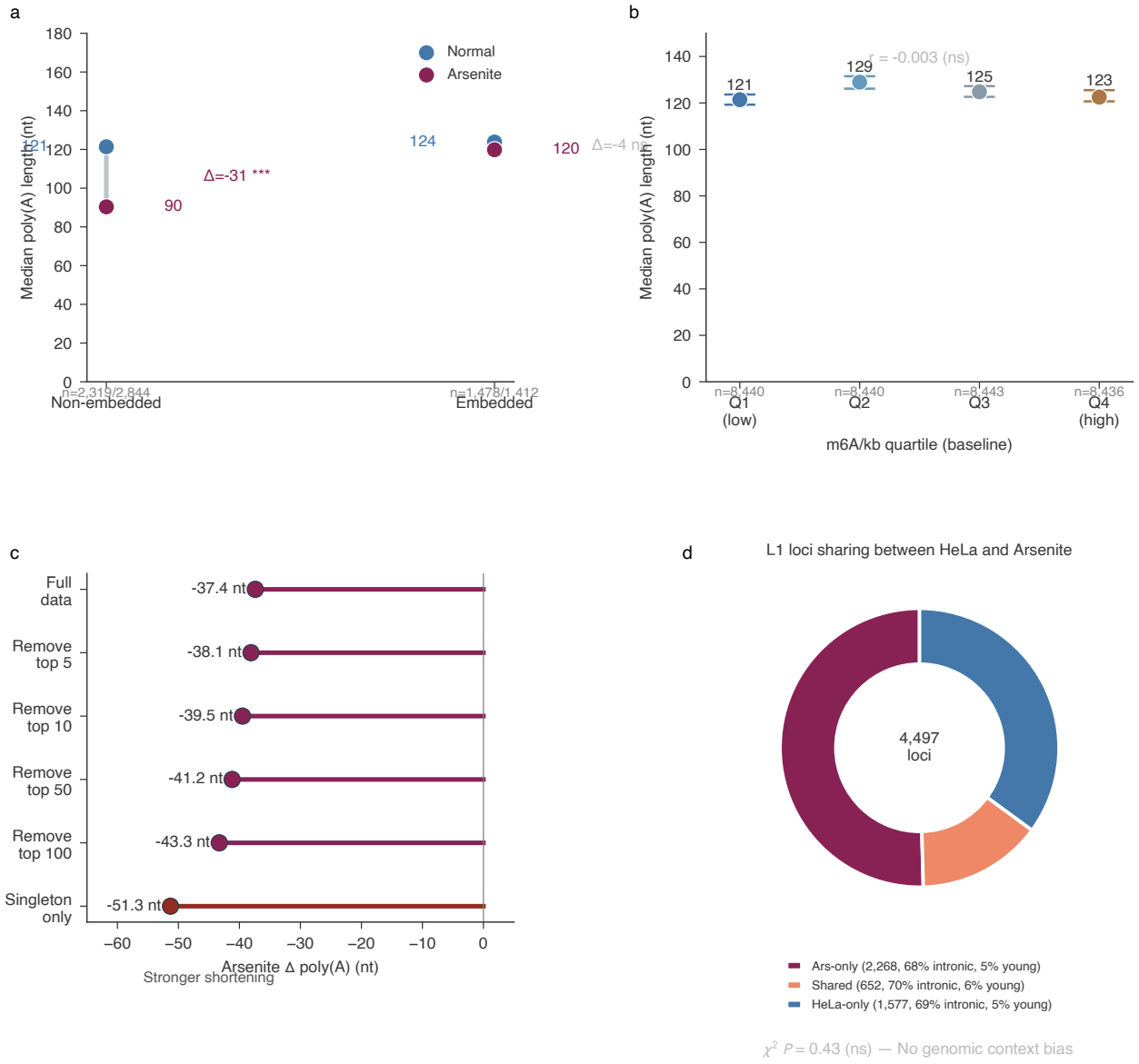

Figure 2: **Exon-embedded L1 are protected and baseline m^{6}A–poly(A) is uncorrelated. (a)** Non-embedded L1 shows arsenite-induced shortening ($\Delta = -31$ nt), while embedded L1 is unaffected ($\Delta = -4$ nt, ns). **(b)** Median poly(A) by m^{6}A/kb quartile under baseline conditions across nine cell lines. The four quartiles are indistinguishable ($r = -0.003$, ns), confirming that the m^{6}A–poly(A) association emerges only under stress.

Supplementary Figure S3: Regulatory L1 Vulnerability

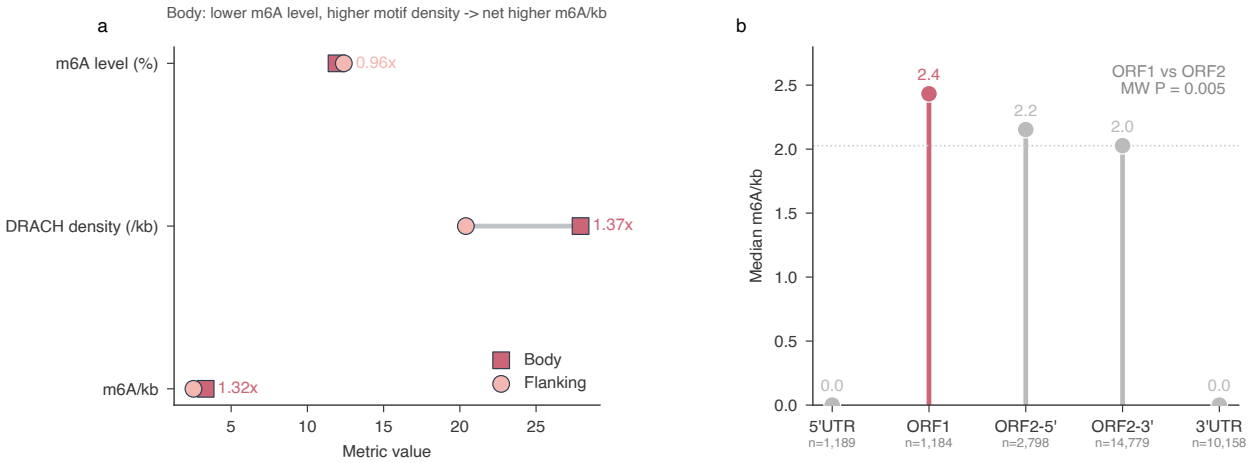

Figure 3: **Regulatory L1 elements are selectively vulnerable to arsenite-induced poly(A) shortening.** **(a)** ECDF of poly(A) for regulatory vs. non-regulatory ancient L1. Regulatory L1 shows $\Delta = -73$ nt under stress, approximately three times the non-regulatory effect ($\Delta = -24$ nt). **(b)** Fraction of reads with poly(A) <30 nt by ChromHMM state. **(c)** m^6A-matched comparison: regulatory L1 is 49–84 nt shorter than non-regulatory at equivalent m^6A levels. **(d)** Within stressed regulatory L1, Q4 (highest m^6A) shows median 59 nt vs. Q1 at 28 nt ($P = 2.9 \times 10^{-4}$).

Supplementary Figure S4: Cross-Cell-Line m^{6}A Consistency

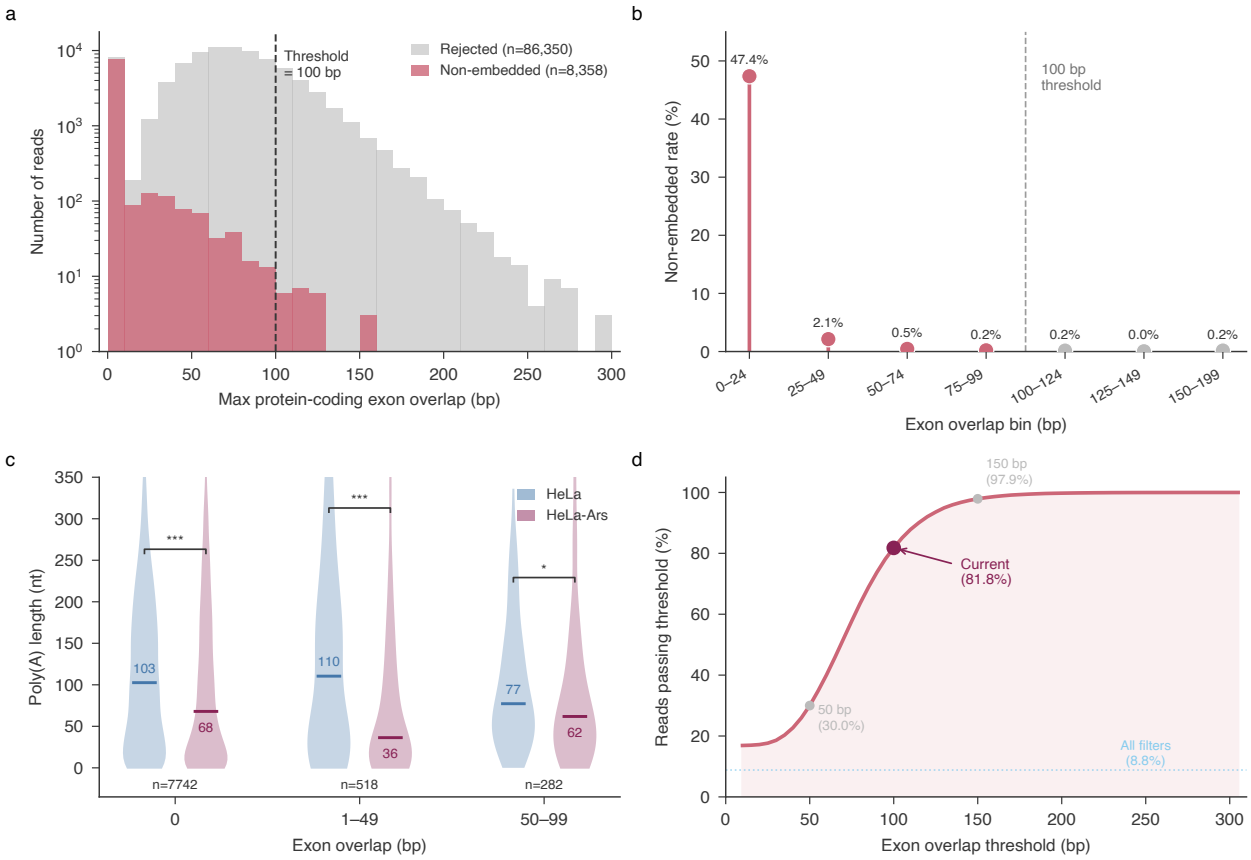

Figure 4: **L1 m^{6}A patterns are consistent across human cell lines. (a)** Median m^{6}A/kb for young and ancient L1 across nine cell lines. All show the Young>Ancient gradient (CV = 0.075). **(b)** Pairwise Spearman correlation of subfamily-level m^{6}A/kb (median $\rho = 0.85$). **(c)** Per-locus consistency (median pairwise $\rho = 0.73$). **(d)** Per-cell-line replicate variation.

Supplementary Figure S5: Illumina RNA-seq Validation

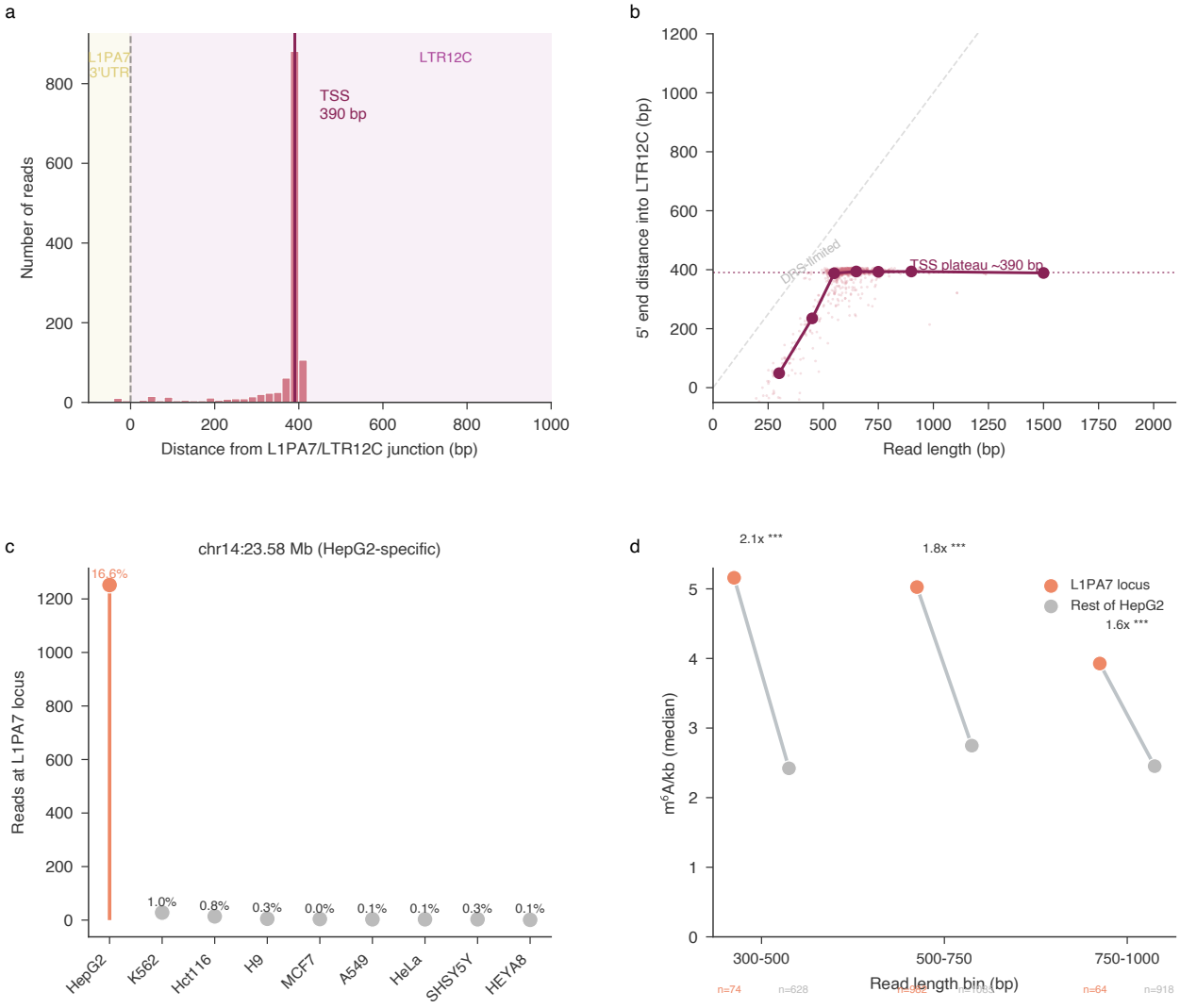

Figure 5: **Illumina RNA-seq confirms net ancient L1 RNA decrease under arsenite stress.** **(a)** Gene-normalized L1 fold change. Ancient L1 show robust net decrease (FC = 0.925). **(b)** Differential expression: 53 of 67 significant ancient L1 subfamilies are downregulated. **(c)** Arsenite treatment validation: canonical stress genes show expected upregulation.

Supplementary Figure S6: Per-Read m^{6}A/kb–Poly(A) Scatter

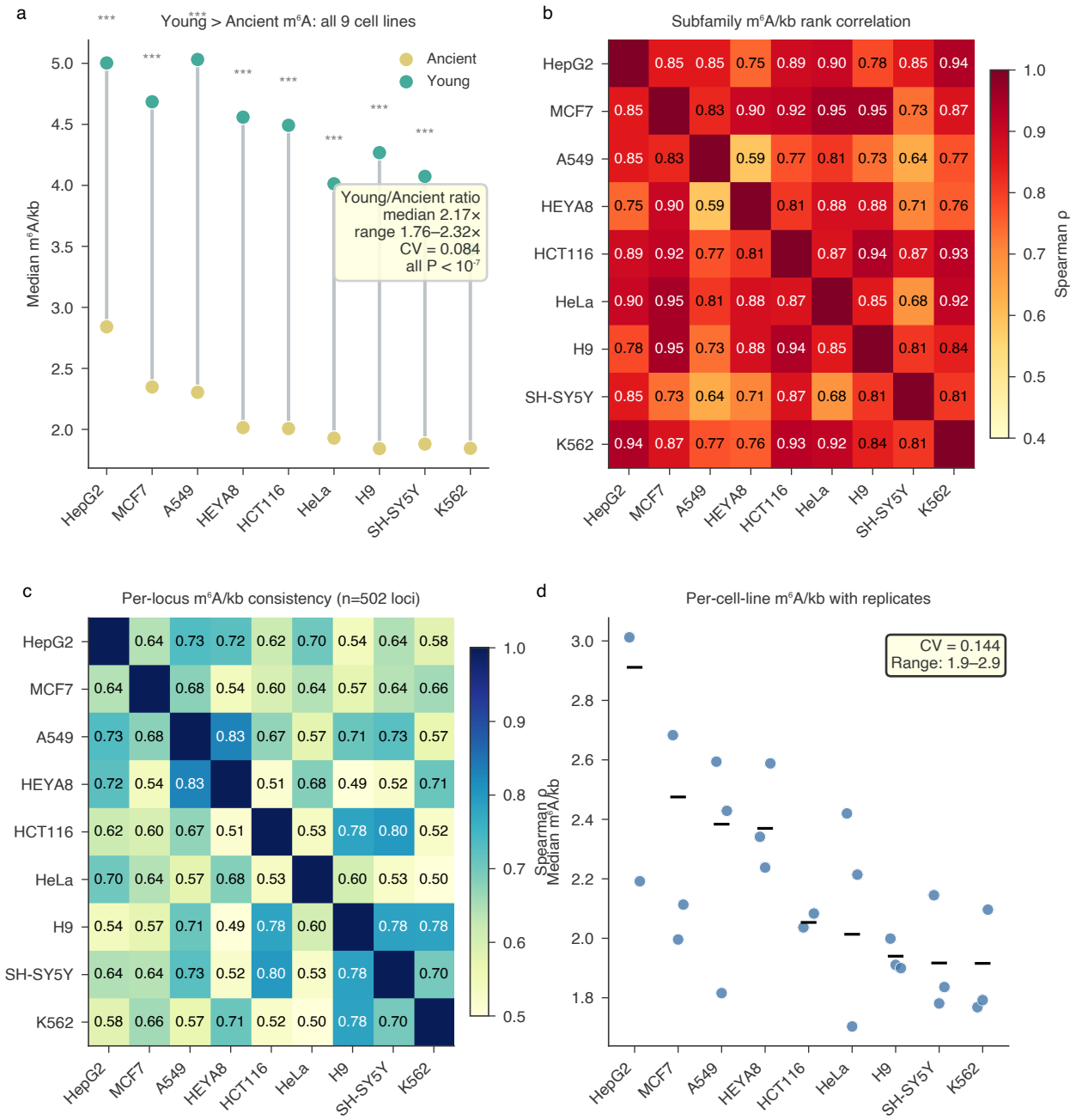

Figure 6: **Per-read m^{6}A/kb versus poly(A) length. (a)** Scatter plots for HeLa (normal) and HeLa-Ars (stress) with regression lines. Under normal conditions the relationship is flat; under stress a positive slope emerges. **(b)** Median Δpoly(A) by m^{6}A/kb quartile. Q1 shows -42 nt shortening, Q4 retains poly(A) (-16 nt, ns). Spearman $\rho = 0.201$, $P = 3.3 \times 10^{-26}$.

Supplementary Figure S7: lncRNA Control

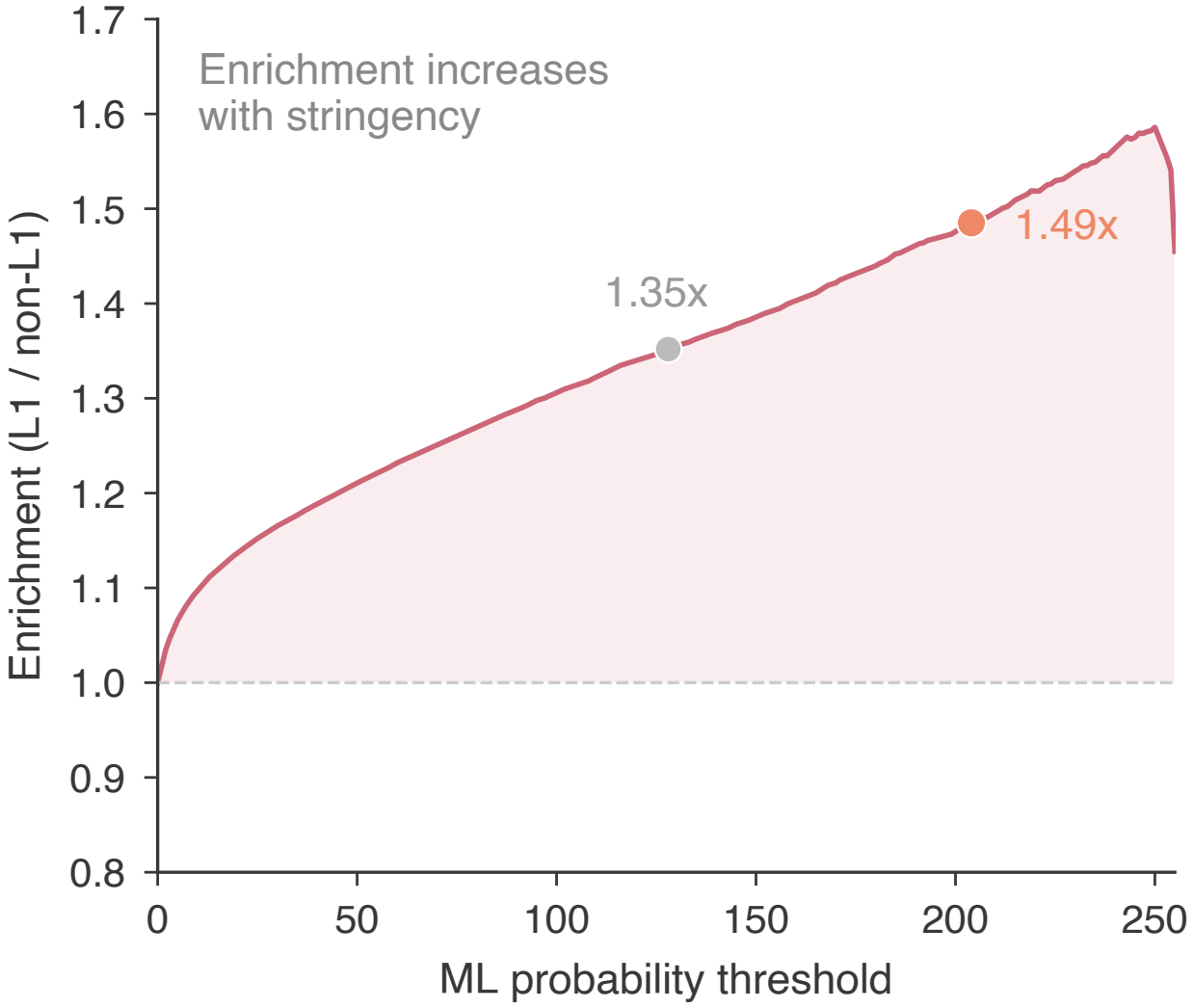

Figure 7: **Annotated lncRNAs do not recapitulate L1-specific poly(A) shortening or m^6A enrichment.** **(a)** Poly(A) distributions: L1 shortens under arsenite ($\Delta = -31$ nt) while lncRNAs are unaffected ($\Delta = -2.9$ nt). **(b)** Median Δpoly(A) across transcript categories. **(c)** m^6A density: L1 carries 1.27-fold higher m^6A than lncRNAs. **(d)** m^6A–poly(A) correlation under stress: L1 $\rho = 0.201$ vs. lncRNA $\rho = 0.065$.

Supplementary Figure S8: Sequence Features of Young L1 Confer Stress Immunity

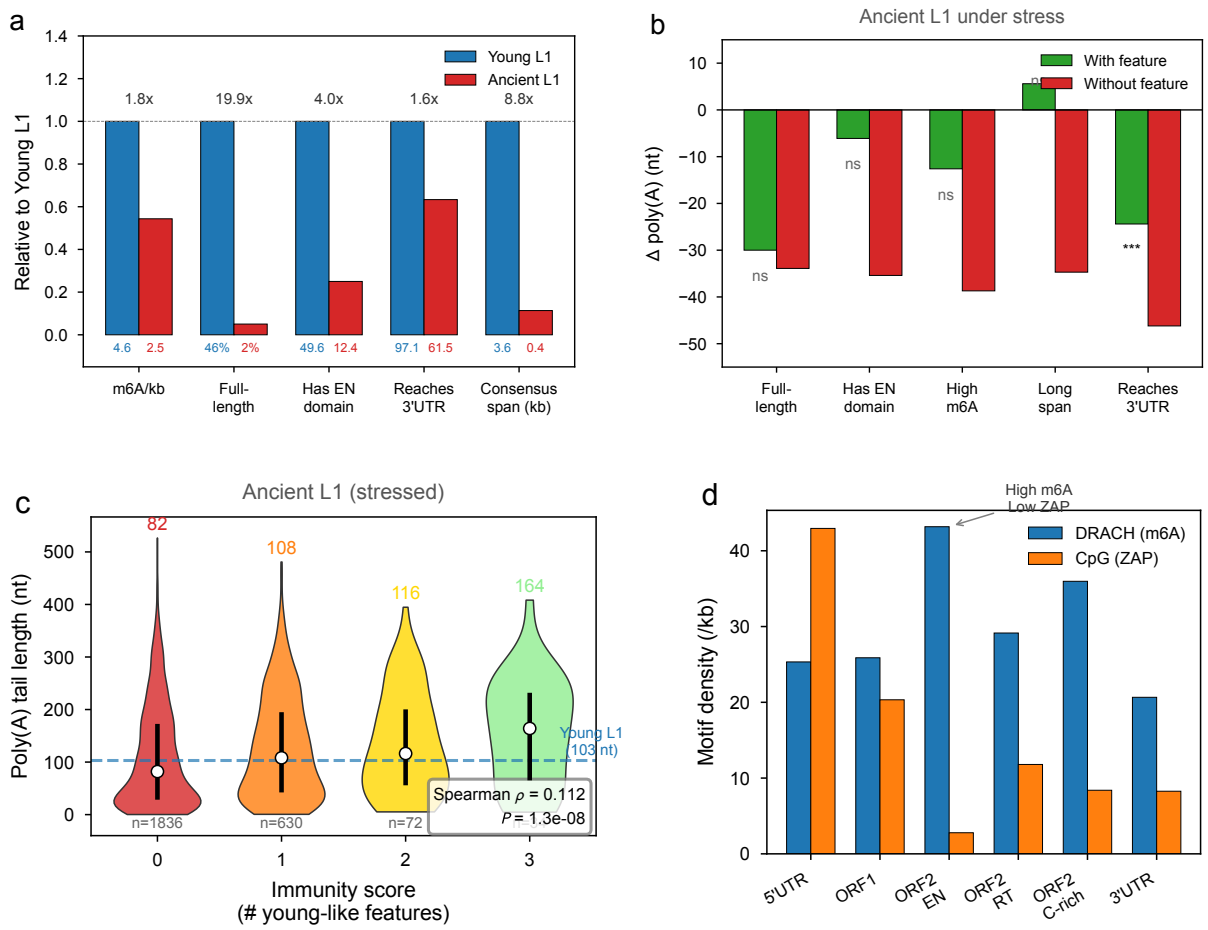

Figure 8: **Sequence features of young L1 confer stress immunity.** **(a)** Young vs. ancient L1 comparison across five immunity-associated features. **(b)** Stress-induced poly(A) change for ancient L1 stratified by presence/absence of each feature. Ancient elements possessing any single feature are immune to shortening. **(c)** Composite immunity score (0–3) predicts poly(A) retention dose-dependently (Spearman $\rho = 0.169$, $P = 5.5 \times 10^{-18}$). **(d)** DRACH and CpG density across L1 functional domains. The EN domain combines the highest DRACH density with near-complete CpG depletion.

Supplementary Tables

Table 1: **Cell line and sequencing summary.** Eleven human cell lines profiled by ONT direct RNA sequencing.

Cell Line	Reps	L1 Reads	Reads/Rep	Loci	Subfamilies	Young (%)	Read Length
MCF7	3	12,545	4,182	4,806	118	7.3	785
HepG2	2	7,552	3,776	2,684	114	4.2	971
A549	3	3,096	1,032	1,686	106	5.1	1,059
HeLa-Ars	3	2,844	948	1,900	107	5.9	935
K562	3	2,827	942	1,810	114	4.5	762
HeLa	3	2,319	773	1,486	105	5.3	1,046
MCF7-EV	1	1,977	1,977	1,130	100	18.2	1,229
Hct116	2	1,653	827	1,025	99	3.3	1,019
H9	3	1,487	496	894	104	5.9	1,277
HEYA8	3	1,161	387	661	97	13.2	1,395
SH-SY5Y	3	1,083	361	757	95	3.1	1,137
Total		**38,544**		**12,125**	**128**	**7.0**	

Table 2: **OLS regression: m^6A–poly(A) coupling model.** Poly(A) tail length as dependent variable (n = 4,970, R^2 = 0.039). The stress×m^6A/kb interaction (bold) indicates stress-specific coupling.

Variable	Coef.	SE	t	P	95% CI
Intercept	128.53	2.97	43.22	$< 10^{-100}$	[122.7, 134.4]
m^6A/kb	2.26	0.85	2.67	7.6×10^{-3}	[0.6, 3.9]
Read length (z)	10.46	1.34	7.80	7.5×10^{-15}	[7.8, 13.1]
Stress	−29.31	3.92	−7.47	9.2×10^{-14}	[−37.0, −21.6]
Young	−17.49	13.31	−1.31	0.189	[−43.6, 8.6]
Stress × m^6A/kb	**4.22**	**1.10**	**3.84**	$\mathbf{1.24 \times 10^{-4}}$	**[2.1, 6.4]**
Young × m^6A/kb	−1.76	2.21	−0.79	0.427	[−6.1, 2.6]
Stress × Young	21.40	12.00	1.78	0.075	[−2.1, 44.9]

References

Table 3: **Threshold robustness of the m^{6}A–poly(A) interaction.** The interaction is significant at all five thresholds tested.

Probability threshold	β_3 (stress×m^{6}A/kb)	SE	P	Model R^2
$\geq$ 50%	3.00	0.74	4.7×10^{-5}	0.040
$\geq$ 60%	3.23	0.82	8.4×10^{-5}	0.039
$\geq$ 70%	3.46	0.93	1.8×10^{-4}	0.039
$\geq$ 80% (primary)	4.22	1.10	1.2×10^{-4}	0.039
$\geq$ 90%	5.52	1.45	1.5×10^{-4}	0.037

Table 4: **Per-locus aggregation increases the m^{6}A–poly(A) effect size.** Per-read baselines (grey) shown for comparison.

Subgroup	Unit	N	Spearman ρ	OLS R^2	WLS R^2	P
All L1, stressed	read	2,733	0.201	0.031	—	1.6×10^{-26}
Ancient, stressed	read	2,534	0.204	0.033	—	3.2×10^{-25}
Ancient, stressed, $\geq$3 reads	locus	136	0.335	0.084	0.124	7.5×10^{-5}
Ancient, stressed, $\geq$5 reads	locus	49	0.513	0.120	0.176	1.7×10^{-4}
Ancient, unstressed, $\geq$3 reads	locus	140	0.039	0.003	—	0.65